Ryan Sheatsley

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Academic & Professional Experience

United States Army Research Laboratory — **Adelphi, MD**
Research Scientist and Team Lead (mentored by Andrew Toth) — *Summers of 2014–2018*

Patent Litigation — **University Park, PA**
Code Review & Document Drafting — *2018–2019*

Systems Internet and Infrastructure Security Laboratory — **University Park, PA**
Systems Administrator — *2015–Present*

Penn State Learning Center — **University Park, PA**
Writing Tutor (mentored by Dr. Jon Olson) — *2012–2013*

Education

Pennsylvania State University — **University Park, PA**
M.S. in Computer Science and Engineering — *2016–2018*

- Thesis: *Adversarial Examples in Constrained Domains*
- Advisor: *Prof. Patrick McDaniel*
- Thesis committee: Prof. Patrick McDaniel, Prof. Trent Jaeger, Prof. Nicolas Papernot

Pennsylvania State University — **University Park, PA**
B.S. in Computer Engineering — *2011–2015*

Publications

Conference Proceedings

Development of Machine Learning Algorithms for Directional Gamma Ray Detection. *Matt Durbin, Ryan Sheatsley, Christopher Balbier, Tristan Grieve, Patrick McDaniel, Azaree Lintereur*. Proceedings of the 60th Annual Meeting of the Institute of Nuclear Materials Management, Palm Sptrings, CA. (2019)

Curie: Policy-based secure data exchange. *Z. Berkay Celik, Abbas Acar, Hidayet Aksu, Ryan Sheatsley, Patrick McDaniel, A Secuk Uluagac*. Proceedings of the Ninth ACM Conference on Data and Application Security and Privacy (CODASPY), Richardson, TX. (2019)

Network Traffic Obfuscation: An Adversarial Machine Learning Approach. *Gunjan Verma, Ertugrul Ciftcioglu, Ryan Sheatsley, Kevin Chain, Lisa Scott*. Proceedings of the 2018 Military Communications Conference (MILCOM), Baltimore, MD. (2018)

Detection under Privileged Information. *Z. Berkay Celik, Patrick McDaniel, Rauf Izmailov, Nicolas Papernot, Ryan Sheatsley, Raquel Alvarez, Ananthram Swami*. Proceedings of the 2018 Asia Conference on Computer and Communications Security (ASIACCS), Incheon, Republic of Korea. (2018)

Heterogeneous information sharing of sensor information in contested environments. *Jason A Wampler, Chien Hsieh, Andrew Toth, Ryan Sheatsley*. Proceedings of Volume 10190 Ground/Air Multisensor Interoperability, Integration, and Networking for Persistent ISR VIII (SPIE), Anaheim, CA. (2017)

Journals

Improving Radioactive Material Localization by Leveraging Cyber-Security Model Optimizations. *Ryan Sheatsley, Matthew Durbin, Azaree Lintereur, and Patrick McDaniel*. IEEE Sensors. (2021)

Book Chapters

Evading Machine Learning-based Network Intrusion Detection Systems with GANs. *Bolor-Erdene Zolbayar, Ryan Sheatsley, Patrick McDaniel*. In: *Charles A Kamhoua, Christopher D. Kiekintveld, Fei Fang, Quanyan Zhu* (Ed.). Game Theory and Machine Learning for Cyber Security. John Wiley & Sons, Incorporated, Hoboken, New Jersey. (2021)

Technical Reports

A Vision Toward an Internet of Battlefield Things (IoBT): Autonomous Classifying Sensor Network. *John Zhu, Egan McClave, Quan Pham, Sujay Polineni, Sam Reinhart, Ryan Sheatsley, Andrew Toth*. (2018)
Cleverhans v2.0.0: an adversarial machine learning library. *Nicolas Papernot, Ian Goodfellow, Ryan Sheatsley, Reuben Feinman, Patrick McDaniel*. (2016)
Analyzing GAIAN Database (GaianDB) on a Tactical Network. *Ryan Sheatsley, Andrew Toth*. (2015)

Thesis

Adversarial Examples in Constrained Domains. *Ryan Sheatsley*. (2018)

Honors

Robert M. Owens Memorial Scholarship: Pennsylvania State University *2017*
Pathways Internship Program: United States Government *2015*
Peer Tutor Faculty Recommendation: Pennsylvania State University *2011*

Talks

Adversarial Examples in Constrained Domains: RIT Great Lakes Security Day *2020*
Adversarial Examples in Constrained Domains: Cyber-CRA Webinar *2020*

Professional Activities

Reviewer

IC: IEEE Transactions on Internet Computing *2021*
TIP: IEEE Transactions on Image Processing *2020*
TEM: IEEE Transactions on Engineering Management *2020*
DKE: Data & Knowledge Engineering Journal *2020*
GLOBECOM: IEEE Global Communications Conference *2019*